

1.2 Classwork Solutions: Speed, Velocity, and Unit Conversion

1. A student walks 24.0 m in 18.5 s.

(a) $v = \frac{d}{t} = \frac{24.0 \text{ m}}{18.5 \text{ s}} = 1.2972 \dots = \mathbf{1.30 \text{ m/s}}$

- (b) **Three significant figures.** Both measurements have three, and a quotient carries no more significant figures than the least precise measurement that went into it. Writing 1.29729 would claim a precision the stopwatch and the tape measure do not have.

2. Convert each speed.

(a) $30.0 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{30,000 \text{ m}}{3600 \text{ s}} = 8.333 \dots = \mathbf{8.33 \text{ m/s}}$

The km cancel and the h cancel, leaving m/s. A useful benchmark: dividing km/h by 3.6 gives m/s.

(b) $25.0 \frac{\text{m}}{\text{s}} \times \frac{1 \text{ km}}{1000 \text{ m}} \times \frac{3600 \text{ s}}{1 \text{ h}} = \mathbf{90.0 \text{ km/h}}$

3. A fastball at 95 mi/h.

(a) $95 \frac{\text{mi}}{\text{h}} \times \frac{5280 \text{ ft}}{1 \text{ mi}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{501,600 \text{ ft}}{3600 \text{ s}} = 139.33 \dots = \mathbf{139 \text{ ft/s}}$

(b) $139.33 \frac{\text{ft}}{\text{s}} \times \frac{1 \text{ m}}{3.28 \text{ ft}} = \mathbf{42.5 \text{ m/s}}$

Either route works. Going through kilometers: $95 \times 1.61 = 153 \text{ km/h}$, and $153 \div 3.6 = 42.5 \text{ m/s}$. Note the full-precision 139.33 was carried into part (b) — rounding to 139 first still gives 42.4, close but not identical. Round only at the end.

4. Put all three in m/s before comparing.

- Car: $60 \frac{\text{mi}}{\text{h}} \times \frac{5280 \text{ ft}}{1 \text{ mi}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 88.0 \text{ ft/s}$, and $88.0 \div 3.28 = \mathbf{26.8 \text{ m/s}}$
- Train: $95 \text{ km/h} \div 3.6 = 26.39 \dots = \mathbf{26.4 \text{ m/s}}$
- Motorcycle: $\mathbf{27 \text{ m/s}}$ (already in m/s)

Slowest to fastest: **train (26.4 m/s) < car (26.8 m/s) < motorcycle (27 m/s)**.

The three are within 3% of each other — the point of the problem is that you cannot rank speeds until they are all in the same unit.

5. Sound travels 343 m/s; the delay is 4.0 s.

$$d = vt = (343 \text{ m/s})(4.0 \text{ s}) = 1372 \text{ m} = \mathbf{1.4 \times 10^3 \text{ m}} \quad (2 \text{ s.f., because } 4.0 \text{ s has two})$$

$$1372 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} \times \frac{1 \text{ mi}}{5280 \text{ ft}} = \mathbf{0.85 \text{ mi}}$$

This is the “count the seconds” rule of thumb: roughly 1 km every 3 seconds, or about 1 mile every 5 seconds.

6. Climbing 5 floors, 18 steps each, 8.0 in per step.

$$(a) \quad 5 \text{ floors} \times \frac{18 \text{ steps}}{1 \text{ floor}} = \mathbf{90 \text{ steps}}$$

$$(b) \quad 90 \text{ steps} \times \frac{8.0 \text{ in}}{1 \text{ step}} = 720 \text{ in}$$

$$720 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}} = \mathbf{60 \text{ ft}} \quad 60 \text{ ft} \times \frac{1 \text{ m}}{3.28 \text{ ft}} = 18.29 \dots = \mathbf{18 \text{ m}}$$

The step counts are exact, so the 8.0 in riser sets the precision: two significant figures.

$$(c) \quad t = \frac{90 \text{ steps}}{1.2 \text{ steps/s}} = \mathbf{75 \text{ s} = 1 \text{ min } 15 \text{ s}}$$

$$(d) \quad v = \frac{60 \text{ ft}}{75 \text{ s}} = 0.80 \frac{\text{ft}}{\text{s}} \times \frac{60 \text{ s}}{1 \text{ min}} = \mathbf{48 \text{ ft/min}}$$

$$v = \frac{18.3 \text{ m}}{75 \text{ s}} = \mathbf{0.24 \text{ m/s}}$$

Check: $0.80 \div 3.28 = 0.244 \text{ m/s}$ — the two routes agree. This is the vertical speed; the distance actually walked along the stairs is longer, since each step also moves you forward.

7. **Challenge.** The conversion itself is right:

$$30 \frac{\text{mi}}{\text{h}} \times \frac{1.61 \text{ km}}{1 \text{ mi}} = 48.3 \text{ km/h}$$

What is wrong is the number of digits. The speedometer reads 30 mi/h — two significant figures — so the converted speed can carry at most two: **48 km/h**. The seven digits in 48.28032 came out of the calculator, not out of the measurement; they claim the car’s speed is known to a hundred-thousandth of a km/h. (The conversion factor is a rounded number too: 1 mi = 1.609344 km.)

Converting a measurement never makes it more precise — it only restates the same precision in a different unit.